

Certified analytical bounds for hypersonic reachable sets

A measure-theoretic profile decomposition in validated interval arithmetic

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Preliminary PhD Research Proposal

October 1, 2026

Abstract

The reachable set of a nonlinear control system can be embedded in a space of occupation measures, where the dynamics become a linear constraint and the reachable set is the union of the supports of the admissible terminal measures. This research asks what that embedding yields in a singular limit, for a vector field known only as a set-valued map enclosed in validated interval arithmetic. On the state space nothing is lost—the family is tight—but its weak limit forgets every event that lasts a vanishing time, and a reachable set is a support, not an integral: the limit satisfies the Liouville equation of the limit dynamics only up to a defect. The target theorem identifies that defect. The loss is read on the energy dissipation along the rescaled time axis, where Lions’ concentration-compactness alternative applies as it stands, and it organizes into a profile decomposition: a limit system in a few slow coordinates, one boundary-layer bubble per concentrating event, entering as a jump, and a remainder. The statement aimed at holds at fixed ε : the reachable set lies in that of the limit hybrid system, thickened by a ball whose radius is an explicit function of ε . Compactness gives the structure and no constants. Those are to come from direct estimates—the energy identity, a matching estimate at each pass, and averaging over the faster oscillations, where frequencies that cross are the critical difficulty—and are required in closed form, in the physical constraint set and the aerodynamic interval widths. Hypersonic entry supplies the instance: a six-degree-of-freedom vehicle whose thin-atmosphere limit furnishes the singular perturbation, whose exo-atmospheric arcs and equilibrium glide furnish the limit system, and whose atmospheric passes furnish the bubbles. The aim is a bound whose dependence on the data is explicit: valid uniformly over the entire admissible family, in the minimal physical state space, and re-evaluated rather than re-solved when the vehicle or the corridor changes.

Contents

1	The analytical problem	4
1.1	Reachable sets as occupation measures	4
1.2	The provenance of the guarantee	5
2	The singular limit and its profile decomposition	6
2.1	Loss of compactness along the time axis and the defect measure	6
2.2	Skeleton, bubbles, residual	7
2.3	Almost-orthogonality of the scaling groups	8
2.4	The certified bound	9
3	The application: hypersonic entry	10
3.1	Why existing methods do not reach it	10
3.2	The state space and the corridor	11
4	Plan of work	12
4.1	Where the work stands	12
4.2	Sequencing	13
A	Coupled entry dynamics	14
A.1	Coordinate rationale	14
A.2	Frames	15
A.3	State vector	15
A.4	Translational kinematics	16
A.5	Gravitational acceleration	16
A.6	Translational dynamics	16
A.7	Rotational dynamics	17
A.8	Aerodynamic force: from quaternion to (F_V, F_γ, F_ψ)	18
A.9	The equilibrium-glide balance	19
A.10	Thermal subsystem (native form)	21
A.11	Ablation	22
A.12	Modal / structural elasticity	22
A.13	Density model and interval envelope	22
A.14	Summary of coordinate choices	22
B	Concentration along the time axis	23
B.1	Setting, and two elementary bounds	23
B.2	Compactness in slow time	24
B.3	The fast-time axis and Lions' lemma	25
B.4	Profiles and their orthogonality	26
B.5	Joint profiles where scales meet	26
B.6	What is not proved	26
C	Skeleton residual and bubble budget	27
C.1	The planar glide	27
C.2	The phugoid	27
C.3	The residual constant	28
C.4	Why this is not a settling rate	28

C.5	Range of validity	29
C.6	The bubble budget	29

1 The analytical problem

The object to be bounded is the reachable set of an atmospheric entry vehicle—every state it can be driven to under admissible control—which is what safety and threat analyses need enclosed. The flight-mechanics specifics are deferred: [Section 3](#) states why entry is the right instance and what its state space looks like, [Appendix A](#) carries the coupled dynamics in full, and [Appendices B and C](#) sketch the limit argument and the constants it rests on. What matters first is that the problem is an analytical one before it is an aerospace one, and that the existing methods fall short of it on two independent counts: the dimension of the state, taken up with the application in [Section 3](#), and the provenance of the guarantee, in [Section 1.2](#).

1.1 Reachable sets as occupation measures

Embed the flight dynamics in a space of measures. Along a trajectory of $\dot{x} = f(x, c, t)$, with the state x in the corridor $\mathcal{X}$ and the control c in $\mathcal{U}$, the time spent in each set defines an occupation measure μ on $[0, T] \times \mathcal{X} \times \mathcal{U}$, and the nonlinear flight mechanics become *linear*: the Liouville equation is a linear constraint tying μ to an initial measure and to a terminal measure μ_T , and the reachable set at time T is the union of $\text{supp } \mu_T$ over the measures that satisfy it. No single measure bounds it. The outer bound comes from the dual: a pair of functions (Φ, Ψ) , the first non-decreasing along every admissible trajectory and at least one on the initial set, the second majorizing it at time T , encloses the reachable set in the superlevel set $\{\Psi \geq 1\}$. This is the occupation-measure program of Lasserre, Henrion and collaborators [\[3, 4\]](#), developed for regions of attraction and invariant sets [\[1, 2\]](#) and, for hybrid systems, for reachable sets and optimal control, where a measure on each guard carries the mass from one mode into the next [\[shia2014convex, zhao2020optimal\]](#). The embedding buys linearity at the price of a relaxation, and the price is known: a measure that satisfies the Liouville constraint is generated by a family of trajectories of the *convexified* inclusion $\dot{x} \in \text{conv } f(x, \mathcal{U}, t)$, not by a single trajectory of the original system [\[1, Lemma 3\]](#). The relaxed reachable set contains the true one, so an outer bound on the first is an outer bound on the second, which is the direction needed here.

The same relaxation admits the interval envelope. Because the true aerodynamics are transcendental and known only to within an envelope, they are enclosed in intervals, and a vector field selected from an interval is a set-valued map; the resulting differential inclusion is the object the measure must describe, not the nominal field [\[6, 7\]](#). Written with the selection as an additional box-valued input, the inclusion is a control system with a bounded disturbance, and at fixed ε its Liouville constraint is the standard one. What is not standard is carrying the outer approximation through the singular limit of [Section 2](#), uniformly over the selections. Separating time scales is itself a standard device of flight mechanics [\[naidu2001singular\]](#), and the classical reduction, which sets $\varepsilon = 0$, requires the fast motion to settle. Occupation-measure theory for inclusions that does not require it already exists, and it is the nearest prior art: it defines an occupational invariant measure without assuming solution uniqueness, which is the same move made here, and lets the slow dynamics be driven by the invariant measures of the fast [\[8\]](#). What it assumes is worth stating plainly, because it locates where the entry problem leaves it. The linear growth condition on the field serves only to exclude finite-time blow-up, and is automatic on a bounded corridor. The singular-perturbation theorem, however, assumes a uniform bound on the fast state along the whole ε -family [\[8, 9\]](#), and the natural fast state of entry is the altitude measured in scale heights: on a corridor of fixed altitude extent it ranges over an interval of length proportional to $1/\varepsilon$, the aerodynamic terms grow exponentially in it, and a skip that rises through a fixed altitude carries it out of every bounded set. The hypothesis holds for motion that stays within a bounded number of

scale heights of the glide, and it fails exactly where a pass concentrates; [Section 2](#) says which is which.

Objective 1: Interval embedding, uniform through the singular limit

Question. What do the occupation measures of the interval inclusion $\dot{x} \in \mathcal{F}_\varepsilon(x, c, t)$ converge to as $\varepsilon \rightarrow 0$, and what linear constraint does the limit satisfy?

To establish. At fixed ε the inclusion is a control system with a box-valued disturbance and its Liouville constraint is standard. So are the bounds that hold uniformly in ε : on mass, on support and, through the energy balance, on dissipation ([Lemma B.2](#)). Two things are not. The first is a certified upper and lower envelope of the atmospheric density, which fixes $\mathcal{F}_\varepsilon$ concretely ([Appendix A.13](#)). The second is the limit of the constraint itself. A weak-* limit of the measures satisfies the Liouville equation of the limit flows only up to a defect, carried where the dissipation concentrates. The claim to be proved is that this defect is a jump term, on a measure supported on the graph of the pass map, so that the limit satisfies the Liouville equation of the hybrid system of [Section 2.4](#), in the form with guard measures [[shia2014convex](#), [zhao2020optimal](#)] (??), uniformly over the selections. Where nothing concentrates the limit is of the kind the existing theory describes, with the oscillation about the glide represented by an invariant measure [\[8\]](#).

Success. The limit constraint stated for the full inclusion, with each hypothesis either verified on the entry corridor or named as an assumption; and its dual, a pair (Φ, Ψ) non-decreasing along the flows and across the jumps of the limit system, as the form in which the outer bound of [Section 1.1](#) survives the limit.

1.2 The provenance of the guarantee

A polynomial barrier whose coefficients come from an interior-point solver has analytical *form* but inherits the solver’s accuracy [\[10\]](#), and the documented failure modes of numerical semidefinite solvers include the loss of strict feasibility on which the repair schemes depend [\[11\]](#). A solver’s output can be made into a certificate: rounded to rational coefficients and re-verified exactly, it is a proof [\[12\]](#). The obstacle is scale, not principle. The rounding needs a strictly feasible Gram matrix, which a tight bound does not have, and exact rational certificates carry a documented bit-complexity whose exponent grows with the number of variables [\[13, 14\]](#)—a cost the thirteen-state field of [Section 3](#) does not invite as a main line.

The program therefore fixes an order of evidence. The certificate itself is a closed-form bound: the constants of [Section 2](#), derived from the decomposition and evaluated over the corridor in validated interval arithmetic. Behind it stand symbolic checks of the derivations those constants rest on, run as part of the build against the same definition of the dynamics that is evaluated numerically. A moment-sum-of-squares computation on a reduced model is the third tier: an independent cross-check, to which the exact-certification machinery above belongs, and never the source of the guarantee.

Interval arithmetic is the setting in which a guarantee has meant, for forty years, an enclosure rather than an approximation. It is rigorous in floating point: every operation is rounded outward, so the computed interval contains the exact result, and nothing is gained for soundness by exact rational arithmetic, which cannot in any case evaluate an exponential or a sine [\[15, Ch. 3\]](#). Two distinct losses have to be managed, and they are often run together. The first is the *dependency problem*. Interval arithmetic is inclusion isotonic—the property that makes an enclosure valid at all—but not every law of real arithmetic survives the move to intervals: distributivity weakens to subdistributivity, and repeated occurrences of a variable are treated as independent [\[15, Chs. 4–5\]](#).

The second is the *wrapping effect*: the image of a box under a flow is not a box, and re-enclosing it at every step admits points that no trajectory reaches [15, Ch. 10]. Refining the step does not help. Under interval Euler steps of size h , re-boxed at each, a rigid rotation inflates a box by $(1 + h)^{2\pi/h}$ per revolution, which *increases* to $e^{2\pi} \approx 535$ as $h \rightarrow 0$; what controls wrapping is the representation of the set—Lohner’s QR factorization, zonotopes, Taylor models [16, 17]. Applied interval analysis supplies the machinery the pipeline is built from, and it is deliberately conservative: certified set inversion, contractors, and validated integration [18].

2.1.2. Symbolic interval propagation

Question. Can the closed-form constants, the pass enclosures they do not cover, and the reachable set of the limit system of Section 2.4 be evaluated over the whole corridor with every rounding error accounted for?

To build. An interval evaluator for the transcendental field and its Jacobian with outward rounding, contractors for the corridor constraints, and a validated integrator for the passes, extending the ball-arithmetic prototype described in Section 4.1. The limit system is itself a reachability problem, in the few slow coordinates and over every bank history: it is to be enclosed by comparison bounds in closed form where they exist—the certified range bound of Section 4.1 is the first—and by validated set propagation where they do not.

Success. Every inequality the certificate asserts is discharged by a program that returns either a proof or an explicit failure to decide at the working precision, and the dynamics it evaluates are the dynamics that were checked symbolically.

2 The singular limit and its profile decomposition

This section carries the new content. The instance supplies a small parameter—in the entry problem the thin-atmosphere ratio $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$, where H_s is the atmospheric scale height and R_e the planetary radius, written ε where the other small parameters are not in play—and the question is what the occupation-measure family does in that limit.

2.1 Loss of compactness along the time axis and the defect measure

On the corridor nothing is lost in the usual sense. The set $[0, T] \times \mathcal{X} \times \mathcal{U}$ is compact, so the occupation measures—which live in the cone $\mathcal{M}_+([0, T] \times \mathcal{X} \times \mathcal{U})$ of nonnegative Radon measures, topologized weakly-* as the dual of $C([0, T] \times \mathcal{X} \times \mathcal{U})$ —form a tight family and converge along subsequences. The limit is nevertheless the wrong object, because it forgets the atmospheric passes. How much it forgets depends on which limit is taken, and that has to be said first.

A pass entered at a flight-path angle γ_e turns the velocity through about $2\gamma_e$ and sheds a share of the kinetic energy of order $\gamma_e/(L/D)$, since lift cannot be had without drag (Eq. C.6). It crosses a scale height in a time $H_s/(V\gamma_e)$, of order $\varepsilon_{\text{atm}}/\gamma_e$ against the slow time, and it levels out where the lift is $V^2\gamma_e^2/(2H_s)$, of order $\gamma_e^2/\varepsilon_{\text{atm}}$ against gravity (Eq. ??). The entry angle therefore enters through $\gamma_e/\sqrt{\varepsilon_{\text{atm}}}$ —as in the classical Chapman–Loh variables, it is $\sqrt{\varepsilon_{\text{atm}}}$ and not ε_{atm} that sets the scale [19, 20]—and two limits have to be told apart. With $\gamma_e/\sqrt{\varepsilon_{\text{atm}}}$ held fixed, a pass lasts a time of order $\sqrt{\varepsilon_{\text{atm}}}$, sheds an energy of the same order, and loads the vehicle by a fixed multiple of what the glide does. This is the limit of a corridor whose dynamic-pressure and load limits are held fixed, and in it nothing concentrates: the passes are many and slight, the troughs of an oscillation about the glide. With γ_e held fixed, a pass lasts a time of order ε_{atm} and sheds a fixed share of the energy, under loads that grow like $1/\varepsilon_{\text{atm}}$. It carries occupation mass of that order, and

none in the limit, while its share of what distinguishes one trajectory from another—the energy dissipated, the heat absorbed, the turn of the velocity vector—does not vanish with it. The physical case lies between the two. $\sqrt{\varepsilon_{\text{atm}}} \approx 0.034$ is about two degrees and entry angles are a few degrees, and a computed pass is neither short against the arc that follows it nor distinct in scale from the oscillation about the glide (Appendix C.6). The statement aimed at is therefore one at fixed ε , with a threshold that decides which passes are resolved one by one, and the two limits are the ends it has to cover.

A reachable set is a support, not an integral, and what has no mass still counts. That alone does not set this problem apart from the limit theory of singularly perturbed control, whose value functions already carry a terminal cost on the slow state [9]. What sets it apart is the hypothesis under which that theory works: the fast state, and the trajectories that approximate an optimum, stay in a bounded set, and a pass that concentrates is what that excludes. The loss has to be read where it occurs, along the fast-time axis $\tau = t/\sqrt{\varepsilon_{\text{atm}}}$, on which the horizon $[0, T/\sqrt{\varepsilon_{\text{atm}}}]$ exhausts the half-line and translations act non-compactly.

Between passes the vehicle is outside the sensible atmosphere, on an arc governed by gravity alone; once captured, it stays near the equilibrium-glide balance $\Lambda = 1$. In the 3-DOF (point-mass) approximation this is the scalar condition $L \cos \sigma / [m(g - V^2/r)] = 1$; in the full 6-DOF formulation the lift component $L \cos \sigma$ is not an independent input but is determined by the attitude quaternion through the body-to-LVLH rotation (Appendices A.8 and A.9), so the balance is a single scalar constraint on the 13-dimensional translational-rotational state, and the slow set is its intersection with the trim and quasi-static-deflection conditions of the two faster groups—one constraint set for each. The balance is taken with planetary rotation and oblate gravity retained (Eq. A.25), and the two do not enter on equal terms: flying east at entry speed the Coriolis term is tens of percent of the net acceleration $g - V^2/r$ that the lift has to balance, while the J_2 terms are a percent of it at most (Remark A.9). The scalar $\Lambda = 1$ is the reduction under spherical gravity, no rotation and small flight-path angle.

The loss is read on a *defect measure*, and the object has to be chosen so that the passes carry mass. The part of the occupation-measure limit that lies off the glide cannot serve, since the passes have none. The energy dissipation can. Along every trajectory of the inclusion the specific mechanical energy in the rotating frame, $\mathcal{E} = \frac{1}{2}V^2 + \Phi_g - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$, changes at the rate $\dot{\mathcal{E}} = -P$, where $P = DV/m \geq 0$ is the drag power: neither the lift nor the Coriolis force does work (Lemma B.2). The total dissipation $\int_0^T P dt \leq \mathcal{E}_{\max} - \mathcal{E}_{\min}$ is therefore bounded uniformly over controls, over selections of the interval field, and over ε . Let

$$\nu_\varepsilon(\tau) = \frac{\sqrt{\varepsilon} P(\sqrt{\varepsilon} \tau)}{\int_0^T P dt} \quad (2.1)$$

be the dissipation density on the fast-time axis, normalized to unit mass and extended by zero. It is a defect measure in the working sense. Where it concentrates is where the limit of the occupation measures stops satisfying the Liouville equation of the limit flow, and the mass it puts there bounds the size of the failure, because the aerodynamic impulse over any interval is bounded by the energy dissipated in it (Lemma B.3). Along any sequence $\varepsilon_n \rightarrow 0$, with a trajectory chosen for each, the ν_ε are nonnegative densities of fixed mass on the line, which is exactly the hypothesis of Lions' concentration-compactness lemma [21, Lemma I.1], and here all three of its alternatives mean something:

- *vanishing*—no window of fixed fast-time width holds a fixed share of the dissipation: the glide and the oscillation about it, on which the drag is bounded and such a window holds a share of order $\sqrt{\varepsilon}$;

- *compactness*, up to translation—one pass carries the dissipation;
- *dichotomy*—the dissipation splits between pieces whose fast-time separation diverges: two passes a positive time apart, or a pass followed by a glide.

Which of the three occurs is decided by the scaling above. With the loads held fixed the drag power is bounded and every sequence vanishes. Concentration needs the fixed entry angle, and then translations along the fast-time axis separate the passes, and it is the dilation half of the principle—the limit-case modification [22]—that resolves each one at its own width, since a pass shorter than $\sqrt{\varepsilon}$ is a Dirac mass at that scale. The trichotomy is Lions’ lemma applied as stated. That the vanishing part is the glide and the concentrated pieces are the passes is asymptotics still to be argued, from the inner and outer expansions of the classical theory [23], and one step of the argument is open (Appendix B).

2.2 Skeleton, bubbles, residual

Rather than restore compactness, the program extracts the profile decomposition that iterating the dichotomy hands it:

- a *skeleton*, low-dimensional—what survives the limit: the exo-atmospheric arcs between passes, which are conics up to the oblateness correction, followed by the equilibrium glide, a graph over (r, V) in the corridor subspace once the balance is solved for the vertical lift fraction. For the glide a rigorous characterization with quantitative accuracy estimates already exists [24], and it treats the glide as a *regular* perturbation: the exact solution is not attracted to the zeroth-order one, a point that returns in Section 2.4;
- one *boundary-layer bubble* per atmospheric pass, the analogue of a concentrating profile: the pass rescaled to its own width. Its net effect on the slow coordinates is a jump. The exit states that the bank history and the selections can produce from one entry state form a compact set, and the map that assigns it, the *pass map* J , is the object to be enclosed. Orthogonality is read off the scale-and-core separation condition that governs concentrating profiles generally [25]. The decomposition of skip entry into an aerodynamic inner region and a gravitational outer one is classical [19, 23], and it has been carried to second order and to three dimensions [kuo1995improved]. Those are expansions of one trajectory, with the error assessed by example. What is not classical is a statement uniform over a family of trajectories, with an explicit remainder at fixed ε ;
- a *residual*: what the finitely many profiles do not capture. In a profile decomposition the remainder is not compact and is not small in the conserved norm; it is small in a weaker one, and only as the number of extracted profiles grows [26, 27]. Here the weaker norm is the largest share of dissipation held by any fast-time window: on a glide the residual carries most of the dissipation, and almost none of it in any one window. The quantitative bound the program needs is a different statement, a distance in state space, and it is taken up in Section 2.4.

The two kinds of statement are kept apart, because they come from different arguments. A decomposition obtained by compactness is a statement about subsequences, and what it gives uniformly it gives by contradiction: it says which pieces can occur, and that nothing else can, and it produces no constant. The wave-equation theory is explicit about this. The uniform bound that follows from its decomposition comes with a constant its authors say they cannot describe [26, Corollary 2]. Here compactness is asked only for the structure. The constants come from direct

estimates: the energy identity, the bound it gives on the aerodynamic impulse, a matching estimate at each pass, and averaging about the glide. Each of those is an inequality with a constant that can be written down, and writing them down is objective 5.

The number of bubbles is bounded *a priori*, and the bound comes from the same monotone quantity. A pass that dissipates at least δ of specific energy can occur at most

$$K_\delta \leq \frac{\mathcal{E}_{\max} - \mathcal{E}_{\min}}{\delta} \quad (2.2)$$

times, for every control and every selection of the interval field. Passes below the threshold are left to the oscillation about the glide and to the residual, which is the device of the profile-decomposition literature: only finitely many profiles exceed any given size. The integrated heat-load budget gives the count its physical reading—a trajectory can only afford so many skips [23, 28]—but it routes through a stagnation-point heating correlation, and the energy budget does not (Remark A.11).

Enclosing a pass is where the wrapping effect has to be met. For linear systems it can be avoided altogether: support-function methods propagate each direction independently, so successive over-approximations do not compound, and what remains is a discretization error of first order in the time step [29]. Nothing comparable is available for a nonlinear pass.

Question. Can the net effect of one atmospheric pass on the slow coordinates be enclosed tightly enough that a sequence of K_δ passes does not exhaust the corridor?

To prove and build. Wrapping in a pass has a specific source. Trajectories of a set enter the layer at different times, the set is sheared along the flow, and re-enclosing the sheared set in a box at every step admits points that no trajectory reaches. The remedy that fits the decomposition is to enclose the *pass map*—entry state to exit state, in the rescaled time of the profile—as a single object, a zonotope or Taylor-model enclosure of a map rather than a tube of boxes, and to compose pass maps with the outer flow.

Success. A bound on the growth of the enclosure per pass, explicit in the interval widths, and its verification on a multi-pass corridor against sampled trajectories.

2.3 Almost-orthogonality of the scaling groups

Three scaling groups act on the time axis. The atmospheric one dilates by $\sqrt{\varepsilon_{\text{atm}}}$; those of the attitude and the elastic motion dilate by $\varepsilon_{\text{att}} = T_{\text{att}}/T_{\text{slow}}$ and $\varepsilon_{\text{ela}} = T_{\text{ela}}/T_{\text{slow}}$, the short-period attitude and first structural periods measured against the slow time $T_{\text{slow}} = \sqrt{R_e/g}$. Their separation is the Bahouri–Gérard almost-orthogonality condition, read on the dissipation: two profiles are orthogonal when their scales diverge or, at equal scale, when their centers separate by a diverging number of widths [26, (1.18)]. Two things follow from it, and they differ in kind. The dissipation is a measure, so it splits *additively* across orthogonal profiles; that is the analogue of the energy identity of the wave-equation theory, and it costs nothing. The state does not split additively, because the profiles interact through the nonlinear dynamics: the state leaving one pass is the datum of the next arc. What has to be proved is the analogue of the nonlinear statement of that theory, that the contributions of separate profiles compose without interacting, up to an error that is explicit and uniform over the family. For passes separated in time this is a matching estimate between inner and outer solutions. For the three groups it is scale separation where the timescale bands are ordered; where two bands cross inside the corridor the groups are not orthogonal there, and the decomposition has to carry one joint profile on the coupled subsystem rather than two. Establishing

this with the scales themselves known only to within the aerodynamic intervals is the one genuinely new obligation, and it sits on the critical path.

Objective 4: Almost-orthogonality under interval disturbances

Question. Do the scaling groups remain orthogonal, in the sense of the Bahouri–Gérard condition, for every selection of the aerodynamic interval field—and where they do not, what replaces the decomposition?

To prove. A non-interaction theorem: for profiles separated in scale or in time, the map from entry data to exit data factors through the individual pass maps and the outer flow, with an explicit remainder, uniformly over selections. The interval widths enter twice: as a disturbance within each profile, and through the periods T_{att} and T_{ela} , which depend on the uncertain coefficients, so that the ordering of the bands is itself a hypothesis to be verified over the corridor. Where two bands cross, the statement is for a joint profile.

Success. The theorem for the atmospheric and attitude groups with the remainder in closed form; the elastic group either included or enclosed as a widened interval, as in the fallback of Section 4.

The argument worth transporting is the constructive one: splitting a bounded sequence into an oscillatory and a singular part at each scale forces those scales to be orthogonal, and iterating extracts a pairwise orthogonal family, so orthogonality is produced by the construction rather than assumed by it [26].

2.4 The certified bound

The target statement is

$$\mathcal{R}_\varepsilon(T) \subseteq \underbrace{\left(\Phi_S(\Theta) \right)}_{\text{skeleton}} \oplus \underbrace{\left(\bigoplus_{j=1}^{K_\delta} \mathcal{I}_j \right)}_{\text{bubbles}} \oplus \underbrace{\mathbb{B}_\varrho(C_{\mathcal{R}}\sqrt{\varepsilon})}_{\text{residual}}, \quad (2.3)$$

uniformly over the admissible family generated by the uncertainty set, not along a nominal trajectory. Here $\mathcal{R}_\varepsilon(T)$ is the set of slow states reachable at time T from the initial set, under every admissible control and every selection of the interval field. Θ is the compact set of skeleton parameters—initial data and slow controls—and $\Phi_S: \Theta \rightarrow \mathcal{X}$ the endpoint map of the skeleton flow. $\mathcal{I}_j$ is the set of displacements of the time- T endpoint attributable to pass j : its impulse set, carried to time T by the skeleton flow, over all states from which the pass can be entered. The sum $\oplus$ is the Minkowski sum, and $\mathbb{B}_\varrho(a)$ is the ball of radius a in the corridor metric ϱ , a weighted supremum norm in which altitude is measured in scale heights and each remaining slow coordinate in units of its corridor width. Passes compose in sequence, not in parallel, and the sum is the telescoped form of that composition: the endpoint of a trajectory with K passes differs from the skeleton endpoint by a sum of K displacements, one from each $\mathcal{I}_j$. The sum forgets that successive displacements are correlated, which makes the bound conservative and not unsound.

The residual constant $C_{\mathcal{R}}$ and the bubble count K_δ are to be derived as explicit, closed-form expressions in the corridor bounds and the aerodynamic interval widths, independent of any numerical optimization. Both exist in first form. The count is Eq. (2.2). On the glide, with L/D the lift-to-drag ratio and σ the bank angle,

$$C_{\mathcal{R}} = \frac{2\sqrt{gR_e}}{V(L/D) \cos \sigma \sqrt{1 - V^2/(gr)}}, \quad (2.4)$$

which bounds the computed distance to the glide to within a few percent over the range of lift-to-drag ratio, speed and bank tested (Appendix C). The same analysis shows what the formula does not yet cover, and the limitation is structural. $C_{\mathcal{R}}\sqrt{\varepsilon}$ is twice the damping ratio of the phugoid, the oscillation of altitude against flight-path angle about the glide. Its damping *rate*, $g/[V(L/D)\cos\sigma]$, does not depend on ε , while its frequency grows like $\varepsilon^{-1/2}$. In the limit the motion normal to the glide is an undamped oscillation on its own timescale: the glide balance is not a normally hyperbolic slow manifold in the sense the geometric theory requires [30], and no boundary-layer estimate of Tikhonov type brings a trajectory onto it. The constant above is the ringing amplitude of a trajectory started level on the glide; a trajectory leaving a pass rings with whatever amplitude the pass left it, for a time comparable to the entry. The residual bound therefore has to carry that amplitude, either as an explicit decaying term or by promoting the phugoid amplitude to a slow coordinate of the skeleton and averaging over its phase; and $C_{\mathcal{R}}$ diverges as V approaches circular speed, which is where the glide gives way to arcs and passes.

In the spherical-LVLH coordinates the altitude and speed bounds of the corridor are box constraints on (r, V) . The heating, dynamic-pressure and load limits are not: each is a curve $h \geq h_{\min}(V)$. Each is, however, monotone in altitude and in speed at fixed attitude, so contracting a box against it returns the exact interval hull of the feasible part, and the interval widths that enter $C_{\mathcal{R}}$ and K_{δ} inherit that tightness (Appendix A.1). Requiring the constants in closed form, rather than as the output of a solver, is what separates a certificate from a computation in the sense of Section 1.2.

Question. What are $C_{\mathcal{R}}$ and K_{δ} as functions of the corridor bounds, the initial set and the aerodynamic interval widths?

To derive. The count from the energy budget, with the threshold δ chosen against the residual it leaves behind. The residual in three parts: the lag behind the glide, which Eq. (2.4) already gives; the ringing left by a pass, which decays at a rate independent of ε and has to be carried explicitly or absorbed into the skeleton by averaging; and the matching error at each pass. Each constant is to hold for every selection of the interval field, and its range of validity—bounded away from circular speed—is part of the statement.

Success. Formulas whose evaluation over the corridor in validated interval arithmetic bounds the distance from sampled trajectories of the full model to skeleton plus bubbles, with no fitted number.

3 The application: hypersonic entry

Entry is not an arbitrary illustration of the machinery above. It supplies the singular perturbation, the skeleton and the concentrating events in one system, and it is a problem the existing methods are known not to reach.

3.1 Why existing methods do not reach it

The direct method solves a Hamilton–Jacobi–Isaacs equation on a grid, which is exponential in state dimension and capped in practice near four or five states [31, 32]. A rigid-body entry vehicle carries thirteen states,

$$x = (r, \lambda, \phi, V, \gamma, \psi, q_0, q_1, q_2, q_3, p, q_b, r_b) \in \mathbb{R}^{13}, \quad (3.1)$$

and $17 + 2N_m$ once integrated heat load, wall temperature, ablation recession, mass loss and N_m aerothermoelastic modes are coupled [33]. The two counts use “degree of freedom” in its two customary senses, and it is worth separating them: the vehicle is six-degree-of-freedom in the configuration sense (three translational, three rotational), while Eq. (3.1) has thirteen coordinates spanning a twelve-dimensional state manifold once the unit-quaternion constraint $\|\vec{q}\| = 1$ is imposed.

So the grid method stops before the real problem starts.

Set-propagation methods go further. Taylor-model and zonotope flowpipes return rigorous outer bounds at ten to twenty states, and they are the baseline this program has to be measured against [17, 34, 35]. What they return is an enclosure of one instance: a numerical set for one vehicle, one initial box and one horizon, whose tightness tends to degrade with the width of the initial set and with the length of weakly damped motion it has to follow—and the phugoid is weakly damped for the whole of the glide. They are the right tool inside a pass, where the horizon is short, and that is how objective 3 uses them. They do not produce a bound whose dependence on the corridor and on the interval widths is explicit, which is what a statement uniform over a family requires.

What makes the measure-theoretic route viable is not that it evades the dimension but that the bound it produces does not scale with it: the skeleton is low-dimensional by construction, the bubbles are compact, and the residual is a single ball.

3.2 The state space and the corridor

The translational state is carried in geocentric spherical position (r, λ, ϕ) and wind-frame velocity (V, γ, ψ) , so that the altitude and speed bounds of the entry corridor are box constraints on states and the heating, dynamic-pressure and load limits are monotone in the two coordinates they bind—the natural setting for interval contraction. Attitude is a unit quaternion, free of Euler-angle singularities. The chart has a price, which is stated rather than hidden: the longitude and heading rates are singular at the geographic poles, so the corridor is taken with $|\phi| \leq \phi_{\max} < \pi/2$, and a trajectory class that overflies a pole needs a second chart or a Cartesian formulation (Appendix A.1).

Because interval arithmetic evaluates transcendentals natively, the algebraic adjunctions that recast the system in polynomial form [36] are not needed on the main line; they survive only inside the moment-sum-of-squares cross-check. The state space therefore carries only the physical dimensions—spherical position, wind-frame velocity, attitude quaternion, body rates, thermal, ablation, and modal states—with no artificial embeddings, and the two auxiliary states of the polynomial formulation, the density root $y = \sqrt{\rho/\rho_0}$ and the wall-temperature surrogate, are gone (Appendix A.10). This is a design decision rather than an open problem, and its trade is stated once: transcendental dynamics in coordinates native to the corridor and the glide, against polynomial dynamics in which the corridor is a nonlinear function of the state (Appendix A.14).

Objective 6: Independent verification

Question. Is the certificate right, by evidence that does not share its derivation?

To build. Three checks, in the order of Section 1.2. Symbolic verification of every derivation behind the constants, run with the build. A moment-sum-of-squares outer approximation on a reduced polynomial model, as a cross-check that shares neither the decomposition nor the interval arithmetic. And a comparison on a common benchmark—a capsule skip entry reconstructed from published data—against Taylor-model flowpipes and against sampled trajectories of the full model, reported for each coordinate group rather than as a single number.

Success. Containment of every sample by the certified set, agreement with the cross-check where both apply, and a stated tightness against the baseline, coordinate by coordinate.

4 Plan of work

Six concrete obligations carry the program; they are stated where each arises above and gathered here, in order, as the plan of work.

1. *Interval embedding, uniform through the singular limit:* the outer approximation for the set-valued (interval) dynamics, with constants independent of ε .
2. *Validated computational pipeline:* interval evaluation, contraction and integration that discharge the certificate with every rounding error accounted for.
3. *Enclosures of the pass maps:* tight enclosures carried through a sequence of atmospheric passes without premature blow-up.
4. *Almost-orthogonality under interval disturbances:* non-interaction of the scaling groups under bounded uncertainty—the **critical path**, on which the profile decomposition and the closed-form bound rest.
5. *Closed-form constants:* $C_{\mathcal{R}}$ and K_{δ} from the physical corridor and the interval widths, free of any numerical solver.
6. *Independent verification:* symbolic checks of the derivations, a moment-sum-of-squares cross-check, and a benchmark against set-propagation tools and sampled trajectories.

Objectives 1–2 establish the interval occupation-measure embedding, 3–4 the profile decomposition, and 5–6 reduce it to closed form and check it; objective 4 gates the rest.

4.1 Where the work stands

The computational layers of the program have working prototypes; the central theorem is open. An implementation of the coupled model is public, with a suite of verification tasks whose failure criteria are stated before they run.¹ Three of its results bear directly on the claims above. The residual scaling has been measured rather than assumed: with the scale height varied over more than a decade, the distance to the glide scales as $\varepsilon^{0.49}$, and the closed form Eq. (2.4) bounds it from above to within about four percent. The translational equations of motion, the energy balance behind Eq. (2.2) and the linearization behind $C_{\mathcal{R}}$ are reduced to identities by computer algebra, against the same definition of the dynamics that the code evaluates in floating point and in interval arithmetic, and compiled numerical routines are generated from those expressions. And a validated-arithmetic

¹<https://github.com/ethank5149/aether>

prototype exists: enclosures of the entry field and reachable tubes in ball arithmetic, with a certified bound on the downrange still available to a capsule on a published skip entry, holding for every bank history that keeps the trajectory inside a stated corridor. None of this is the theorem. It is evidence that the constants are computable and that the pipeline can be built, and the same suite records the obstacle of [Section 2.4](#), the weakly damped phugoid, as a computed fact rather than a conjecture.

4.2 Sequencing

The dependency structure suggests three phases rather than six parallel tracks. The first establishes the embedding—objective 1, together with the certified envelope of the atmospheric density that it needs in order to be stated on a concrete inclusion ([Appendix A.13](#)). The second attacks the critical path: objective 4, which is where the program either works or does not, and which should therefore be reached early enough that a negative result is still recoverable. Objective 3 runs alongside it, since the almost-orthogonality argument and the enclosure geometry constrain each other. The third phase reduces the decomposition to closed form—objective 5—completes the validated pipeline of objective 2, and runs the symbolic and moment-sum-of-squares checks of objective 6 against the result. The pipeline is the only deliverable that cannot be shortened by a weaker theorem.

Two retreat positions are worth identifying now. If objective 4 resists a proof for all three scaling groups simultaneously, the fallback is to establish it for the atmospheric and attitude groups and to treat the elastic group as a widened interval rather than a separate profile. That weakens the residual constant without invalidating the decomposition. And if the ringing left by a pass cannot be bounded in closed form, the phugoid amplitude is carried as one more slow coordinate of the skeleton, which costs a dimension and keeps the bound.

A Coupled entry dynamics

The identities displayed in this appendix—the gravity components, the equations of motion and their classical reductions, the quaternion kinematics, the glide balance—are checked by computer algebra. For the translational equations the check is made against the same definition of the field that the accompanying code evaluates in floating point and in interval arithmetic, and it derives them from Newton’s law in the rotating frame rather than taking them on trust from the literature.

A.1 Coordinate rationale

The coordinate choice is dictated by two competing requirements of the interval-arithmetic pipeline:

1. **The corridor must contract tightly.** The entry corridor—bounded altitude, bounded speed, and limits on heating rate, dynamic pressure and load factor—defines the region whose reachable subset we certify. In spherical position and wind-frame speed the first two are box constraints on states, $h_{\min} \leq r - R_e \leq h_{\max}$ and $V_{\min} \leq V \leq V_{\max}$, for which the interval hull is exact. The other three are not boxes: each is a curve $h \geq h_{\min}(V)$ in the altitude–speed plane, the load limit depending on attitude as well. But the dynamic pressure $\frac{1}{2}\rho V^2$ and the stagnation heating rate of Eq. (A.26) are each decreasing in altitude and increasing in speed, so the part of a box that satisfies the limit has an interval hull that is computed exactly, by solving the limit along two edges of the box, with no dependency loss. What a box cannot represent is the curved boundary itself, and that is recovered by subdivision.
2. **The glide balance must be native.** The equilibrium-glide condition $\Lambda = 1$ relates altitude, speed and the vertical component of the lift; to leading order it does not involve the flight-path angle. Solved for the vertical lift fraction—the bank angle, in the point-mass model—it is a graph over (r, V) , on which the flight-path angle is determined by (r, V) at the next order. The profile decomposition—skeleton, bubbles, residual—is organized around this surface. Were it a derived nonlinear function of the states (as it would be in Cartesian position and velocity), every evaluation of the distance to it would pay a dependency loss.

The translational state is therefore carried in spherical-LVLH coordinates $(r, \lambda, \phi, V, \gamma, \psi)$. Attitude is carried as a unit quaternion $\vec{q} = (q_0, q_1, q_2, q_3)^T$ relating the body frame $\mathcal{B}$ and the ECEF frame $\mathcal{E}$ (its direction-cosine matrix $\mathbf{R}_{\mathcal{E}\mathcal{B}} = \mathbf{R}(\vec{q})$ is given in Eq. (A.1))—this avoids all Euler-angle singularities and keeps the quaternion kinematics in their simplest form.

The chart is singular at the geographic poles $\phi = \pm\pi/2$, and in two places: the longitude rate Eq. (A.4), through $1/\cos\phi$, and the heading rate Eq. (A.10), through $\tan\phi$. Both are coordinate singularities, not physical ones, but only the first is harmless. Longitude is cyclic in this model—it enters neither the gravity field, the corridor constraints nor the glide condition—whereas heading enters the Coriolis and oblateness terms and cannot be integrated separately. The corridor is therefore taken with $|\phi| \leq \phi_{\max} < \pi/2$. That is a restriction on the trajectory class and is stated as one: a ground track that overflies a pole needs a second chart, or the Cartesian formulation of Appendix A.14. Re-orienting the coordinate system to move the pole away from the trajectory is not available, because the gravity field and the rotation vector are axisymmetric about the geographic axis and no other: in a tilted chart the eastward gravity component no longer vanishes and $\vec{\omega}_E$ acquires components along all three axes.

The Savageau–Voit polynomial recasting of the thermal subsystem is not used. Because the interval pipeline evaluates transcendentals natively, the density-square-root embedding $y = \sqrt{\rho/\rho_0}$ and the polynomial surrogate for wall temperature are unnecessary, and the raw physical variables

are retained. The recasting survives only inside the moment-sum-of-squares cross-check, which needs polynomial data.

A.2 Frames

Rotation convention. Throughout, $\mathbf{R}_{\mathcal{AB}}$ denotes the rotation that carries coordinates *from* frame $\mathcal{B}$ *to* frame $\mathcal{A}$, i.e. $\vec{v}^{\mathcal{A}} = \mathbf{R}_{\mathcal{AB}} \vec{v}^{\mathcal{B}}$; its columns are the axes of $\mathcal{B}$ expressed in $\mathcal{A}$. Each such matrix is orthogonal, so $\mathbf{R}_{\mathcal{AB}}^{-1} = \mathbf{R}_{\mathcal{AB}}^{\top} = \mathbf{R}_{\mathcal{BA}}$, and subscripts compose in the usual way, $\mathbf{R}_{\mathcal{AB}}\mathbf{R}_{\mathcal{BC}} = \mathbf{R}_{\mathcal{AC}}$.

- $\mathcal{E}$ (**ECEF**): rotating with the constant planetary rate $\vec{\omega}_E = \omega_E \hat{z}_{\mathcal{E}}$.
- $\mathcal{L}$ (**LVLH / NED**): at geocentric position (r, λ, ϕ) , with basis vectors

$$\begin{aligned}\hat{e}_N &= -\sin \phi \cos \lambda \hat{x}_{\mathcal{E}} - \sin \phi \sin \lambda \hat{y}_{\mathcal{E}} + \cos \phi \hat{z}_{\mathcal{E}}, \\ \hat{e}_E &= -\sin \lambda \hat{x}_{\mathcal{E}} + \cos \lambda \hat{y}_{\mathcal{E}}, \\ \hat{e}_D &= -\cos \phi \cos \lambda \hat{x}_{\mathcal{E}} - \cos \phi \sin \lambda \hat{y}_{\mathcal{E}} - \sin \phi \hat{z}_{\mathcal{E}}.\end{aligned}$$

The rotation $\mathbf{R}_{\mathcal{EL}}$ from $\mathcal{L}$ to $\mathcal{E}$ (columns are the above vectors in $\mathcal{E}$) is a function of (λ, ϕ) alone.

- $\mathcal{W}$ (**Wind frame**): x -axis along the planet-relative velocity, third axis in the $\vec{V}-\hat{e}_D$ plane, y -axis completing the right-handed system. The triad actually used below is $(\hat{V}, \hat{n}_{\gamma}, \hat{n}_{\psi})$ of Eqs. (A.18) to (A.20), in which $\hat{n}_{\gamma}$ points along increasing flight-path angle—radially *outward* at level flight, $\hat{n}_{\gamma}|_{\gamma=0} = -\hat{e}_D$ —so that a positive normal force appears as a positive F_{γ} and the point-mass identification $F_{\gamma} = L \cos \sigma$ carries the expected sign. Parameterized by flight-path angle γ (positive climbing) and heading ψ (clockwise from North):

$$\hat{V} = \cos \gamma \cos \psi \hat{e}_N + \cos \gamma \sin \psi \hat{e}_E - \sin \gamma \hat{e}_D.$$

- $\mathcal{B}$ (**Body frame**): fixed at the instantaneous centre of mass. Related to $\mathcal{E}$ by the attitude quaternion $\vec{q}$ through the direction-cosine matrix $\mathbf{R}_{\mathcal{EB}} = \mathbf{R}(\vec{q})$, which carries body components to $\mathcal{E}$ ($\vec{v}^{\mathcal{E}} = \mathbf{R}(\vec{q}) \vec{v}^{\mathcal{B}}$):

$$\mathbf{R}(\vec{q}) = \begin{pmatrix} q_0^2 + q_1^2 - q_2^2 - q_3^2 & 2(q_1 q_2 - q_0 q_3) & 2(q_1 q_3 + q_0 q_2) \\ 2(q_1 q_2 + q_0 q_3) & q_0^2 - q_1^2 + q_2^2 - q_3^2 & 2(q_2 q_3 - q_0 q_1) \\ 2(q_1 q_3 - q_0 q_2) & 2(q_2 q_3 + q_0 q_1) & q_0^2 - q_1^2 - q_2^2 + q_3^2 \end{pmatrix}, \quad (\text{A.1})$$

quadratic in $\vec{q}$ (its columns are the body axes expressed in $\mathcal{E}$).

A.3 State vector

The full coupled state is

$$x = \underbrace{(r, \lambda, \phi, V, \gamma, \psi)}_6, \underbrace{(q_0, q_1, q_2, q_3, p, q_b, r_b)}_7, \underbrace{(Q_{\text{tot}}, T_w, s, m)}_4, \underbrace{(\eta_1, \dots, \eta_{N_m}, \dot{\eta}_1, \dots, \dot{\eta}_{N_m})}_{2N_m}, \quad (\text{A.2})$$

with the unit-quaternion constraint $\|\vec{q}\| = 1$ making the translational-rotational block effectively 12-dimensional. Total dimension: $17 + 2N_m$.

A.4 Translational kinematics

The planet-relative velocity in the $\mathcal{L}$ (NED) frame is $\vec{V}_{\mathcal{L}} = (V \cos \gamma \cos \psi, V \cos \gamma \sin \psi, -V \sin \gamma)^\top$. The kinematic equations are:

$$\dot{r} = V \sin \gamma, \quad (\text{A.3})$$

$$\dot{\lambda} = \frac{V \cos \gamma \sin \psi}{r \cos \phi}, \quad (\text{A.4})$$

$$\dot{\phi} = \frac{V \cos \gamma \cos \psi}{r}. \quad (\text{A.5})$$

A.5 Gravitational acceleration

The J_2 gravitational potential in geocentric spherical coordinates is

$$\Phi_g(r, \phi) = -\frac{\mu}{r} \left[1 - \frac{1}{2} J_2 \left(\frac{R_e}{r} \right)^2 (3 \sin^2 \phi - 1) \right].$$

Its gradient, resolved in $\mathcal{L}$ (NED), has only radial and northward components:

$$g_D(r, \phi) = \frac{\mu}{r^2} \left[1 + \frac{3}{2} J_2 \left(\frac{R_e}{r} \right)^2 (1 - 3 \sin^2 \phi) \right], \quad (\text{A.6})$$

$$g_N(r, \phi) = -\frac{3\mu}{r^2} J_2 \left(\frac{R_e}{r} \right)^2 \sin \phi \cos \phi, \quad (\text{A.7})$$

$$g_E = 0 \quad (\text{by axial symmetry}).$$

Remark A.1. This reproduces the Cartesian form under the identifications $r_3/\|\vec{r}\| = \sin \phi$ and $\|\vec{r}\| = r$, but the reduction is not a term-by-term match and is worth doing explicitly. The Cartesian J_2 perturbation carries the factor $(1 - 5(r_3/\|\vec{r}\|)^2)$ on its two equatorial components and $(3 - 5(r_3/\|\vec{r}\|)^2)$ on its polar one; projecting that vector onto $\hat{r}$ and using $\cos^2 \phi + \sin^2 \phi = 1$ collapses the pair to

$$(1 - 5 \sin^2 \phi) \cos^2 \phi + (3 - 5 \sin^2 \phi) \sin^2 \phi = 1 - 3 \sin^2 \phi,$$

which is the factor appearing in [Eq. \(A.6\)](#). Carrying the Cartesian $(1 - 5 \sin^2 \phi)$ into the radial component instead would double the polar correction and move its sign change from $\phi = \arcsin(1/\sqrt{3}) \approx 35.3^\circ$ to $\arcsin(1/\sqrt{5}) \approx 26.6^\circ$. The spherical form makes the interval evaluation tighter: g_D and g_N are explicit functions of (r, ϕ) , with no square root of a sum of squares.

A.6 Translational dynamics

Newton's law in the rotating ECEF frame, resolved along the wind-frame axes (tangential $\hat{V}$, normal-in-vertical-plane, lateral), gives the force equations. Define the aerodynamic force components:

$$\begin{aligned} F_V &= \text{force along } \hat{V}, \\ F_\gamma &= \text{force } \perp \hat{V}, \text{ in the vertical plane,} \\ F_\psi &= \text{force } \perp \text{ the vertical plane (lateral).} \end{aligned}$$

In the 3-DOF model $F_V = -D$, $F_\gamma = L \cos \sigma$, $F_\psi = L \sin \sigma$. In the full 6-DOF formulation these are derived from the attitude quaternion—see [Appendix A.8](#).

The translational dynamical equations are:

$$\dot{V} = \frac{F_V}{m} - g_D \sin \gamma + g_N \cos \gamma \cos \psi + \omega_E^2 r \cos \phi [\sin \gamma \cos \phi - \cos \gamma \cos \psi \sin \phi], \quad (\text{A.8})$$

$$\dot{\gamma} = \frac{1}{V} \left\{ \frac{F_\gamma}{m} - \left(g_D - \frac{V^2}{r} \right) \cos \gamma - g_N \sin \gamma \cos \psi + 2\omega_E V \cos \phi \sin \psi + \omega_E^2 r \cos \phi [\cos \gamma \cos \phi + \sin \gamma \cos \psi \sin \phi] \right\}, \quad (\text{A.9})$$

$$\dot{\psi} = \frac{1}{V \cos \gamma} \left\{ \frac{F_\psi}{m} + \frac{V^2}{r} \cos^2 \gamma \sin \psi \tan \phi - g_N \sin \psi + 2\omega_E V [\sin \phi \cos \gamma - \sin \gamma \cos \phi \cos \psi] + \omega_E^2 r \cos \phi \sin \phi \sin \psi \right\}. \quad (\text{A.10})$$

Remark A.2. With $g_N = 0$ (spherical gravity) and $\omega_E = 0$ (non-rotating planet), these reduce to the classical Vinh equations for unpowered entry [37]:

$$\dot{V} = \frac{F_V}{m} - g \sin \gamma, \quad V \dot{\gamma} = \frac{F_\gamma}{m} - \left(g - \frac{V^2}{r} \right) \cos \gamma, \quad V \cos \gamma \dot{\psi} = \frac{F_\psi}{m} + \frac{V^2}{r} \cos^2 \gamma \sin \psi \tan \phi.$$

This reduction is the working check on the signs in Eqs. (A.8) to (A.10), and it is worth performing rather than assuming: the three equations are assembled by projecting Newton's law onto the wind triad Eqs. (A.18) to (A.20), and the transport term $\vec{\omega}_{\mathcal{LE}} \times \vec{V}$ contributes to each projection with a different power of $\cos \gamma$.

Remark A.3. The sign of the curvature term in Eq. (A.10) is fixed by a one-line test. Take spherical gravity, a non-rotating planet, no lateral aerodynamic force, and level flight ($\gamma = 0$) at northern latitude $\phi > 0$ on a due-east heading ($\psi = \pi/2$). Then $\dot{\psi} = (V/r) \tan \phi > 0$: with ψ measured clockwise from North, an increasing ψ turns the velocity from east toward south, which is the equatorward bending of a great circle. The opposite sign would curve the ground track poleward, and because the J_2 heading drift of Remark A.10 is read off this same equation, a sign error here would propagate into the bubble dynamics rather than cancel.

Remark A.4. The factor $1/\cos \gamma$ in $\dot{\psi}$ is singular in vertical flight ($\gamma = \pm\pi/2$), which the corridor excludes: flight-path angles in lifting entry are a few degrees, and at the apex of a skip $\gamma = 0$. For the interval pipeline the factor is enclosed by interval evaluation of $1/\cos[\gamma]$ provided the interval $[\gamma]$ does not contain $\pm\pi/2$, and the same holds for $1/\cos[\phi]$ and $\tan[\phi]$ under the polar restriction of Appendix A.1.

A.7 Rotational dynamics

The state carries $\vec{\omega} = (p, q_b, r_b)^\top$, the angular velocity of $\mathcal{B}$ with respect to the *inertial* frame, resolved in body-frame components; this is the vector for which Euler's equations Eq. (A.14) take their standard form. The quaternion of Appendix A.1 relates $\mathcal{B}$ to the *rotating* frame $\mathcal{E}$, so its kinematics are driven by the body rate relative to $\mathcal{E}$,

$$\vec{\omega}_{\mathcal{BE}} = \vec{\omega} - \mathbf{R}(\vec{q})^\top \vec{\omega}_E, \quad (\text{A.11})$$

and not by $\vec{\omega}$ itself:

$$\dot{\vec{q}} = \frac{1}{2} \mathbf{\Omega}(\vec{\omega}_{\mathcal{BE}}) \vec{q}, \quad (\text{A.12})$$

with

$$\mathbf{\Omega}(\vec{a}) = \begin{pmatrix} 0 & -a_1 & -a_2 & -a_3 \\ a_1 & 0 & a_3 & -a_2 \\ a_2 & -a_3 & 0 & a_1 \\ a_3 & a_2 & -a_1 & 0 \end{pmatrix}. \quad (\text{A.13})$$

Euler's equations, with inertia tensor $\mathbf{I} = \mathbf{I}(s, \eta)$ depending on recession depth and modal amplitudes:

$$\mathbf{I} \dot{\vec{\omega}} = \vec{M} - \vec{\omega} \times (\mathbf{I} \vec{\omega}) - \dot{\mathbf{I}} \vec{\omega}, \quad (\text{A.14})$$

where $\vec{M} = \vec{M}_a + \vec{M}_{\text{rcs}}$ is the total external moment in $\mathcal{B}$.

Remark A.5. Equation (A.11) is the one place where the choice of a rotating base frame reaches the attitude state, and it cannot be dropped as a small term. Omitting it leaves Eq. (A.12) driven by the inertial rate, which biases the attitude by $\|\vec{\omega}_E\| \approx 4.18 \times 10^{-3} \text{ deg/s}$ —about 6° over a 1500s multi-skip entry. The bias matters more than its size suggests. It is systematic rather than bounded-uncertain, so interval arithmetic does not absorb it: it translates the enclosure instead of widening it, and an enclosure translated off the true attitude is no longer an enclosure. Because the glide balance Eq. (A.25) constrains a specific component of $\mathbf{R}_{\mathcal{LB}} \vec{F}_a^{\mathcal{B}}$, the same offset enters the glide balance directly, and the skeleton would be extracted about the wrong surface.

A.8 Aerodynamic force: from quaternion to (F_V, F_γ, F_ψ)

This section is the bridge between the rotational and translational dynamics.

Step 1: Body-frame aerodynamic force. Given angle of attack α , sideslip β , dynamic pressure $q_{\text{dyn}} = \frac{1}{2} \rho V^2$, and reference area S :

$$\vec{F}_a^{\mathcal{B}} = q_{\text{dyn}} S \left(-C_A(\alpha, \beta, M), C_Y(\alpha, \beta, M), -C_N(\alpha, \beta, M) \right)^\top. \quad (\text{A.15})$$

Step 2: Transform to LVLH. Define the body-to-LVLH rotation

$$\mathbf{R}_{\mathcal{LB}}(\vec{q}, \lambda, \phi) := \mathbf{R}_{\mathcal{EL}}^\top(\lambda, \phi) \mathbf{R}(\vec{q}) = \mathbf{R}_{\mathcal{LE}} \mathbf{R}_{\mathcal{EB}}, \quad (\text{A.16})$$

so that $\vec{v}^{\mathcal{L}} = \mathbf{R}_{\mathcal{LB}} \vec{v}^{\mathcal{B}}$. Then the aerodynamic force in $\mathcal{L}$ is

$$\vec{F}_a^{\mathcal{L}} = \mathbf{R}_{\mathcal{LB}} \vec{F}_a^{\mathcal{B}}. \quad (\text{A.17})$$

Step 3: Project onto wind axes. The wind-axis triad, expressed in $\mathcal{L}$, is

$$\hat{V}_{\mathcal{L}} = (\cos \gamma \cos \psi, \cos \gamma \sin \psi, -\sin \gamma)^\top, \quad (\text{A.18})$$

$$\hat{n}_\gamma = (-\sin \gamma \cos \psi, -\sin \gamma \sin \psi, -\cos \gamma)^\top, \quad (\text{A.19})$$

$$\hat{n}_\psi = \hat{V}_{\mathcal{L}} \times \hat{n}_\gamma. \quad (\text{A.20})$$

Then:

$$F_V = \vec{F}_a^{\mathcal{L}} \cdot \hat{V}_{\mathcal{L}}, \quad F_\gamma = \vec{F}_a^{\mathcal{L}} \cdot \hat{n}_\gamma, \quad F_\psi = \vec{F}_a^{\mathcal{L}} \cdot \hat{n}_\psi. \quad (\text{A.21})$$

Step 4: Identification with lift, drag, bank angle. In the point-mass limit where the attitude is in trim, $F_V = -D$, $F_\gamma = L \cos \sigma$, $F_\psi = L \sin \sigma$. In the full 6-DOF formulation D , L , σ are *not independent inputs*: they are determined by the attitude quaternion $\vec{q}$, the aerodynamic coefficients, and the velocity direction (γ, ψ) through Eqs. (A.15)–(A.21).

Remark A.6. The effective bank angle can be extracted as $\sigma = \text{atan2}(F_\psi, F_\gamma)$, an algebraic function of $(\vec{q}, \lambda, \phi, \gamma, \psi, \alpha, \beta)$. Under interval arithmetic, σ is evaluated as an interval atan2 , a standard operation in validated numerics [15].

Remark A.7. The angle of attack α and sideslip β are determined by the quaternion and velocity direction. Let $\vec{V}^B = \mathbf{R}_{\mathcal{L}\mathcal{B}}^\top \vec{V}_\mathcal{L} = (V_x^B, V_y^B, V_z^B)^\top$ be the vehicle's velocity in body-frame components. Then

$$\alpha = \text{atan2}(V_z^B, V_x^B), \quad \beta = \arcsin(V_y^B/V). \quad (\text{A.22})$$

Both are referred to the same vector, which is the only condition that has to be checked and the easiest one to lose: the freestream direction is $-\vec{V}^B$, and defining α from the freestream while defining β from the velocity offsets α by π , so that purely axial flight $\vec{V}^B = (V, 0, 0)^\top$ returns $\alpha = 180^\circ$ instead of 0. Equation (A.22) fixes the velocity as the reference for both; the equivalent freestream convention is $\alpha = \text{atan2}(-V_z^B, -V_x^B)$ together with $\beta = \arcsin(-V_y^B/V)$. The distinction is not cosmetic, because α and β are the arguments of the aerodynamic coefficients in Eq. (A.15): a π offset evaluates the vehicle reversed in its own aerodynamic database, and under interval arithmetic the resulting enclosure is wide and wrong rather than visibly broken.

A.9 The equilibrium-glide balance

The equilibrium-glide condition is the balance about which the translational dynamics organize under the thin-atmosphere singular perturbation $\varepsilon_{\text{atm}} = H_s/R_e \rightarrow 0$. It is the set on which the flight-path angle holds, and Appendix C takes up in what sense, and how weakly, trajectories are drawn to it.

Setting $\dot{\gamma} = 0$ in Eq. (A.9), with every term kept:

$$\begin{aligned} \frac{F_\gamma}{m} = \left(g_D - \frac{V^2}{r} \right) \cos \gamma + g_N \sin \gamma \cos \psi - 2\omega_E V \cos \phi \sin \psi \\ - \omega_E^2 r \cos \phi [\cos \gamma \cos \phi + \sin \gamma \cos \psi \sin \phi]. \end{aligned} \quad (\text{A.23})$$

For small flight-path angle, spherical gravity ($g_N = 0$, $g_D = \mu/r^2$) and no rotation this is the classical condition

$$\frac{F_\gamma}{m} = g - \frac{V^2}{r}, \quad \text{i.e.} \quad \frac{L \cos \sigma}{m(g - V^2/r)} = 1 \quad (\Lambda = 1). \quad (\text{A.24})$$

In the full 6-DOF formulation, F_γ is determined by the quaternion via Eqs. (A.15)–(A.21). The balance therefore constrains the quaternion directly:

$$\frac{1}{m} [\mathbf{R}_{\mathcal{L}\mathcal{B}} \vec{F}_a^B] \cdot \hat{n}_\gamma = \text{right-hand side of Eq. (A.23)}, \quad (\text{A.25})$$

a single scalar constraint on the coupled translational and rotational state (dimension 13); it defines a codimension-1 set. The skeleton of Section 2 is its intersection with the corresponding conditions for the two faster groups: aerodynamic trim, on which the moment of Eq. (A.14) vanishes, and quasi-static deflection, on which each modal amplitude of Eq. (A.29) follows its forcing.

Remark A.8. The glide balance is *not* a constraint on σ alone—it constrains a specific component of $\mathbf{R}_{\mathcal{L}\mathcal{B}}\vec{F}_a^{\mathcal{B}}$, which depends on the full quaternion and on the aerodynamic angles (α, β) implicitly. The “bank angle” that appears in the point-mass glide condition is a derived quantity, and the balance constrains the attitude directly.

Remark A.9. Equation (A.23) retains planetary rotation and oblateness, and the two do not enter on equal terms. Level and due east at an altitude of 50 km, the terms the vertical lift has to balance are, in m/s^2 :

	$g - V^2/r$	Coriolis (equator)	centrifugal (equator)	J_2 , radial (equator to 80°)
$V = 7 \text{ km/s}$	2.04	1.02	0.034	+0.015 to -0.029
$V = 5 \text{ km/s}$	5.77	0.73	0.034	+0.015 to -0.029
$V = 3 \text{ km/s}$	8.26	0.44	0.034	+0.015 to -0.029

The Coriolis term is half of the balance at 7 km/s and an eighth of it at 5 km/s; the centrifugal and J_2 terms are each under two percent of it. The Coriolis term also depends on heading: it vanishes for flight along a meridian and reverses sign westbound, so with rotation retained the glide altitude is a function of heading as well as of speed. A glide condition that kept J_2 and dropped rotation would keep the smaller correction, which is why Eq. (A.23) is written with both.

The classical form $\Lambda = L \cos \sigma / [m(g - V^2/r)] = 1$ is the reduction under spherical gravity, no rotation and small flight-path angle. The J_2 correction to it acts through the $(1 - 3 \sin^2 \phi)$ factor of Eq. (A.6): it decreases monotonically with latitude and *changes sign* partway, being $+0.16\%$ of the leading gravitational term at the equator, zero at $\phi = \arcsin(1/\sqrt{3}) \approx 35.3^\circ$, and -0.32% at the poles. A trajectory crossing 35.3° has the correction act in both directions on a single pass. For the interval pipeline it is enclosed as it stands.

Remark A.10. What makes the J_2 terms worth retaining is not their size but their sign structure over a ground track that spans many degrees of latitude. During a multi-skip entry two J_2 effects act, and they have *different* sign structures—which is the substantive point, since it is what forbids treating either as a monotone accumulation.

1. The northward gravitational component $g_N(r, \phi)$ enters the heading rate $\dot{\psi}$ (Eq. A.10), producing a latitude-dependent heading drift during each skip that is absent under spherical gravity. Its shape is $-\sin \phi \cos \phi$: odd in ϕ , so the sign is fixed by the hemisphere; vanishing at the equator and at the poles; largest at $|\phi| = 45^\circ$, where it reaches 0.16% of the leading gravitational term. Within one hemisphere the drift therefore keeps a single sign and *does* accumulate across successive skips. Northward and southward skips are not symmetric. The Coriolis term of the same equation, $2\omega_E \sin \phi$ in level flight, has the same hemisphere-fixed sign and is twenty to fifty times larger over the range of entry speeds, so the J_2 drift is a correction to a rotation-driven one and not the leading effect.
2. The radial component $g_D(r, \phi)$ modifies the effective centrifugal balance $g_D - V^2/r$ that governs the re-entry angle after each skip peak. Its shape is $(1 - 3 \sin^2 \phi)$: even in ϕ , so it acts identically in both hemispheres, and it changes sign at $|\phi| = \arcsin(1/\sqrt{3}) \approx 35.3^\circ$, running from $+0.16\%$ of the leading term at the equator to -0.32% at the poles. A ground track confined to one side of 35.3° thus accumulates this correction with a fixed sign, while one that crosses 35.3° has it partially cancel between passes.

The consequence for the certificate is a prohibition rather than an estimate. Over K skips the ground track may span a wide latitude range, and the two effects are evaluated at a different ϕ at

every re-entry; because their sign structures differ, no single worst-case latitude bounds both at once. The enclosure may not be tightened by assuming the corrections cancel, and need not be loosened by assuming they reinforce: both are carried term by term. For the interval pipeline this happens automatically: the interval evaluations of $g_D([r], [\phi])$ and $g_N([r], [\phi])$ widen with the latitude interval, which is itself widest during a skip, where the ground track sweeps the most latitude—and an interval $[\phi]$ straddling 35.3° correctly returns a g_D correction of both signs rather than a spuriously signed one. Retaining J_2 costs the interval pipeline nothing and removes a systematic bias of a few tenths of a percent. It is the rotation terms, not J_2 , that the leading-order balance cannot do without.

A.10 Thermal subsystem (native form)

The atmosphere density profile is $\rho(h)$ from the NRLMSIS 2.0 model [38], with $h = r - R_e$. Under interval arithmetic, $\rho(h)$ is enclosed by evaluating the model’s analytic approximation (or a validated piecewise-exponential envelope) with interval argument $[h]$.

Stagnation-point heat flux (Sutton–Graves [39], within its stated validity envelope: axisymmetric blunt bodies, chemical equilibrium, enthalpy 2.3–116.2 MJ/kg, pressure 0.001–100 atm), with Q_{tot} the heat load it accumulates per unit area:

$$\dot{Q}_{\text{tot}} = \dot{q}_s = k_Q \sqrt{\rho/R_n} V^3. \quad (\text{A.26})$$

Wall temperature, for a lumped surface layer of areal heat capacity C_w that re-radiates with emissivity ϵ_w and conducts into a structure at temperature T_b on a timescale τ_{cond} :

$$\dot{T}_w = \frac{\dot{q}_s - \epsilon_w \sigma_{\text{SB}} T_w^4}{C_w} - \frac{T_w - T_b}{\tau_{\text{cond}}}, \quad (\text{A.27})$$

with σ_{SB} the Stefan–Boltzmann constant. No embedding, no recasting. The transcendentals $\sqrt{\rho}$, V^3 and T_w^4 are evaluated directly as interval operations, and each is monotone in its argument.

Remark A.11. Equation (A.26) is used for the corridor constraints and for the thermal states, and deliberately not for the bubble count. Bounding the number of passes by the integrated heat-load budget, $K \leq K_{\text{max}}(Q_{\text{max}})$, would make the count depend on a heating correlation, and in the unsafe direction: an upper bound on K needs a certified *lower* bound on the heat absorbed in each pass, and an equilibrium correlation does not supply one, since heat transfer to a non-catalytic wall from a non-equilibrium boundary layer falls below the equilibrium value [40]. The energy budget of Eq. (2.2) removes the dependence. The count is bounded by the dissipation, which is a property of the dynamics and of the drag enclosure alone.

What remains is the ordinary obligation to use the correlation inside its envelope. The heat load of a pass is accumulated near its lowest point, where the density is highest and the equilibrium assumption is at its best; the high-altitude part of a pass, where that assumption is weakest, contributes little. Where the corridor reaches conditions outside the envelope, the coefficient k_Q is widened to an interval or the correlation is replaced, and the switch is enclosed as a hybrid guard.

Remark A.12. The density-square-root embedding $y = \sqrt{\rho/\rho_0}$ of the original formulation was introduced to render the heating correlation polynomial in the states (the Savageau–Voit recasting strategy [36]). Since the interval pipeline evaluates $\sqrt{\cdot}$ natively, this auxiliary state is eliminated, and with it the dependency loss of carrying an extra, constrained coordinate.

A.11 Ablation

Recession rate coupled to heat flux; mass loss from char-layer blowoff:

$$\dot{s} = \alpha_s \dot{Q}_{\text{tot}}, \quad \dot{m} = -\dot{m}_{\text{abl}}. \quad (\text{A.28})$$

A.12 Modal / structural elasticity

Per mode i , with piston-theory forcing and temperature-dependent natural frequency:

$$\ddot{\eta}_i + 2\zeta_i \omega_i(T_w) \dot{\eta}_i + \omega_i(T_w)^2 \eta_i = Q_i(x, \eta, \dot{\eta}). \quad (\text{A.29})$$

Piston theory is a double expansion in thickness ratio and inverse Mach number valid when the contributions of order δ/M^3 and δ^2/M^2 are negligible [41, 42]; its applicability conditions must be checked at each atmospheric pass. At the bottom of a pass, at high angle of attack and with strong shock interaction, piston theory may fail, and the modal forcing is then carried as uncertain (a widened interval).

A.13 Density model and interval envelope

The functional form of the atmospheric density profile suitable for interval evaluation is not yet settled, and it is the first thing Objective 1 has to fix. The NRLMSIS 2.0 model provides the reference; the interval pipeline requires either a closed-form analytic approximation with certified error bounds, or a piecewise-exponential envelope

$$\rho_{\text{lower}}(h) \leq \rho(h) \leq \rho_{\text{upper}}(h)$$

valid across the altitude range of the corridor. The width $\rho_{\text{upper}} - \rho_{\text{lower}}$ at each altitude is the primary driver of the aerodynamic interval widths.

A.14 Summary of coordinate choices

Feature	Spherical-LVLH + Quat.	Cartesian ECEF + Quat.
Corridor constraints	Box in (r, V) ; monotone limits	Nonlinear: $h = \ \vec{r}\ - R_e$
Glide condition	Native: Eq. (A.24)/(A.23)	Derived from (u, v, w) , $\ \vec{r}\ $
Chart singularity	Poles, in $\dot{\lambda}$ and $\dot{\psi}$	None
Dynamics	Transcendental (sin, cos)	Polynomial (quadratic in $\vec{q}$)
Dependency loss	Trigonometric factors	Corridor and glide evaluation
Density evaluation	$\rho(r - R_e)$: direct	$\rho(\ \vec{r}\ - R_e)$: extra $\sqrt{\cdot}$

The spherical-LVLH choice places the dependency loss in the trigonometric factors of the dynamics, each a function of a single state that encloses tightly. The Cartesian choice places it in the corridor and the glide balance, which become nonlinear functions of several states and are evaluated at every contraction. Neither choice touches the wrapping effect proper, which is a property of how the propagated set is represented and is not reduced by refining the time step: re-boxed at every step, a rotation inflates a box by a factor that tends to $e^{2\pi}$ per revolution as the step vanishes, and the phugoid of Appendix C is such a rotation. For the profile decomposition—which is organized around the corridor and the glide—the spherical choice is the better fit, at the price of the polar restriction of Appendix A.1.

B Concentration along the time axis

This appendix supports [Section 2](#). It is a sketch, and it says where it stops. Two elementary bounds and the compactness they give are proved; the application of Lions' lemma is a statement about a sequence of densities and is exact; the identification of its alternatives with the flight regimes, and the passage from the dissipation back to the state, are stated as what remains to be shown, and gathered in [Appendix B.6](#).

B.1 Setting, and two elementary bounds

Throughout, $\varepsilon = \varepsilon_{\text{atm}}$. The corridor $\mathcal{X}$ is a compact set of states on which $V \geq V_{\min} > 0$, $|\gamma| \leq \gamma_{\max} < \pi/2$ and $|\phi| \leq \phi_{\max} < \pi/2$. A *trajectory* is an absolutely continuous $x_\varepsilon: [0, T] \rightarrow \mathcal{X}$ whose translational part satisfies [Eqs. \(A.3\) to \(A.5\)](#) and [Eqs. \(A.8\) to \(A.10\)](#) for almost every t , with the specific aerodynamic force $(F_V, F_\gamma, F_\psi)/m$ selected measurably from the enclosure. Nothing is assumed about how the force arises: the bank history, the attitude and the selection from the interval field are all free. One assumption is made on the enclosure.

Assumption B.1 (Dissipative enclosure). There is a constant $E_{\max}$ such that every selection satisfies $F_V \leq 0$ and $(F_\gamma^2 + F_\psi^2)^{1/2} \leq E_{\max}|F_V|$.

The first inequality says that the aerodynamic force never does positive work on the vehicle, the second that lift is bounded by a multiple of drag. For a lift-and-drag enclosure they read $[C_D] \subset (0, \infty)$ and $[C_L]/[C_D] \subset [-E_{\max}, E_{\max}]$. A propulsive mode adds a bounded source to what follows and is not considered here.

Lemma B.2 (Energy balance). *Let*

$$\mathcal{E} = \frac{1}{2}V^2 + \Phi_g(r, \phi) - \frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$$

be the specific mechanical energy in the rotating frame, with Φ_g the potential of [Appendix A.5](#). Along every trajectory

$$\dot{\mathcal{E}} = \frac{F_V}{m} V =: -P \leq 0, \quad (\text{B.1})$$

so that $\int_0^T P dt = \mathcal{E}(x(0)) - \mathcal{E}(x(T)) \leq \Delta\mathcal{E}$, where $\Delta\mathcal{E} = \max_{\mathcal{X}} \mathcal{E} - \min_{\mathcal{X}} \mathcal{E}$.

Proof. Differentiate along [Eqs. \(A.3\) to \(A.5\)](#) and [Eq. \(A.8\)](#). The terms in g_D and g_N cancel against $\partial_r \Phi_g \dot{r} + \partial_\phi \Phi_g \dot{\phi}$, by the definition of g_D and g_N as the gradient of Φ_g , and the centrifugal term of [Eq. \(A.8\)](#) against the derivative of $\frac{1}{2}\omega_E^2 r^2 \cos^2 \phi$. The Coriolis force and the two normal components of the aerodynamic force are perpendicular to the velocity and do not appear in [Eq. \(A.8\)](#) at all. The identity is checked by computer algebra with oblateness and rotation retained. $\square$

Lemma B.3 (Impulse bound). *Under [Assumption B.1](#) the aerodynamic acceleration $\vec{a} = \vec{F}_a/m$ satisfies $|\vec{a}| \leq C_a P$ with $C_a = \sqrt{1 + E_{\max}^2} / V_{\min}$. Hence for every interval $I \subseteq [0, T]$,*

$$\int_I |\vec{a}| dt \leq C_a \int_I P dt.$$

Proof. $|\vec{a}|^2 = (F_V^2 + F_\gamma^2 + F_\psi^2)/m^2 \leq (1 + E_{\max}^2) F_V^2/m^2$ and $|F_V|/m = P/V \leq P/V_{\min}$. $\square$

The second lemma is what makes the dissipation the right quantity to follow. Lift cannot be had without drag, so wherever no energy is lost the aerodynamics does nothing, and the motion is that of the gravity field in the rotating frame. Whatever a pass does to the state, the dissipation sees it.

B.2 Compactness in slow time

Let $\varepsilon_n \rightarrow 0$ and let x_n be a trajectory for each n . Write $\mathcal{E}_n(t) = \mathcal{E}(x_n(t))$; $\pi_n = P_n dt$, a positive measure on $[0, T]$ of mass at most $\Delta\mathcal{E}$; and $\vec{y}_n(t), \vec{v}_n(t) \in \mathbb{R}^3$ for the position and the planet-relative velocity in planet-fixed Cartesian components, in which the chart singularities play no part.

Proposition B.4 (Skeleton and impulses as a limit). *Under [Assumption B.1](#) there is a subsequence along which:*

- (i) $\mathcal{E}_n(t) \rightarrow \mathcal{E}_0(t)$ for every $t \in [0, T]$, with $\mathcal{E}_0$ non-increasing, and $\pi_n \rightharpoonup \pi_0$ weakly-*, where π_0 is the measure with distribution function $\mathcal{E}_0(0) - \mathcal{E}_0$;
- (ii) $\pi_0 = p dt + \sum_j a_j \delta_{t_j} + \pi_c$, with $p \in L^1(0, T)$, at most countably many atoms $a_j > 0$, and π_c singular and without atoms; and for every $\delta > 0$

$$\#\{j : a_j \geq \delta\} \leq \Delta\mathcal{E}/\delta;$$

- (iii) on every closed interval on which $\mathcal{E}_0$ is continuous, the convergence $\mathcal{E}_n \rightarrow \mathcal{E}_0$ is uniform;
- (iv) $\vec{y}_n \rightarrow \vec{y}_0$ uniformly, $\vec{v}_n(t) \rightarrow \vec{v}_0(t)$ for every t , $\dot{\vec{y}}_0 = \vec{v}_0$ almost everywhere, and for $0 \leq s < t \leq T$

$$|\vec{v}_0(t) - \vec{v}_0(s)| \leq C_g(t - s) + C_a \pi_0([s, t]), \quad (\text{B.2})$$

with C_g a bound on the gravitational, Coriolis and centrifugal accelerations over the corridor.

Proof. (i) The functions $\mathcal{E}_n$ are non-increasing by [Lemma B.2](#) and uniformly bounded on the compact corridor, so Helly's selection theorem gives a subsequence converging at every point; the measures π_n are their Stieltjes measures, and pointwise convergence of the distribution functions at the continuity points of the limit, with bounded total mass, is weak-* convergence. (ii) is the Lebesgue decomposition of π_0 , and the count follows from $\sum_j a_j \leq \pi_0([0, T]) \leq \Delta\mathcal{E}$. (iii) is Pólya's theorem: monotone functions converging pointwise to a continuous limit on a compact interval converge uniformly. (iv) In planet-fixed components Newton's law reads $\dot{\vec{v}}_n = \vec{a}_n + \vec{g}(\vec{y}_n) - 2\vec{\omega}_E \times \vec{v}_n - \vec{\omega}_E \times (\vec{\omega}_E \times \vec{y}_n)$, so by [Lemma B.3](#)

$$|\vec{v}_n(t) - \vec{v}_n(s)| \leq C_g(t - s) + C_a \pi_n([s, t]).$$

The velocities therefore have total variation at most $C_g T + C_a \Delta\mathcal{E}$, uniformly in n , and Helly's theorem for functions of bounded variation gives pointwise convergence along a further subsequence. The positions are uniformly Lipschitz and converge uniformly by Arzelà–Ascoli, and $\dot{\vec{y}}_0 = \vec{v}_0$ by dominated convergence. [Equation \(B.2\)](#) follows on letting $n \rightarrow \infty$, since $\limsup_n \pi_n(F) \leq \pi_0(F)$ for closed F . $\square$

Three things can be read off [Eq. \(B.2\)](#). The limiting velocity has bounded variation, and it can jump only at an atom of π_0 , by at most $C_a a_j$: an atom is an impulse, and its size bounds the impulse. On an interval that π_0 does not charge, $\int |\vec{a}_n| dt \rightarrow 0$ and the limit is an arc of the gravity field alone. And where π_0 has a density p the aerodynamic forcing survives as an acceleration of magnitude at most $C_a p$, which is the glide. This is the skeleton-and-impulses picture as a statement about limits, valid for every control and every selection, and it is as far as compactness alone goes. It does not exclude the part π_c , and it says nothing yet about what happens *inside* an atom.

B.3 The fast-time axis and Lions' lemma

To see inside an atom the time axis is dilated. Fix a scale $h_n \rightarrow 0$ —in the body, $h_n = \sqrt{\varepsilon_n}$ —suppose the total dissipation stays bounded below, $\pi_n([0, T]) \geq m_0 > 0$, and set

$$\nu_n(\tau) = \frac{h_n P_n(h_n \tau)}{\pi_n([0, T])} \quad (\tau \in \mathbb{R}),$$

extended by zero outside $[0, T/h_n]$. Then $\nu_n \geq 0$ and $\int_{\mathbb{R}} \nu_n d\tau = 1$, and the concentration-compactness lemma [21, Lemma I.1] applies with no further hypothesis. There is a subsequence along which one of the following holds.

Compactness. There are $\tau_n \in \mathbb{R}$ such that for every $\eta > 0$ there is $R < \infty$ with $\int_{\tau_n - R}^{\tau_n + R} \nu_n d\tau \geq 1 - \eta$ for all n .

Vanishing. For every $R < \infty$, $\sup_{y \in \mathbb{R}} \int_{y-R}^{y+R} \nu_n d\tau \rightarrow 0$.

Dichotomy. There is $\alpha \in (0, 1)$ such that for every $\eta > 0$ there are nonnegative ν_n^1, ν_n^2 with $\|\nu_n - \nu_n^1 - \nu_n^2\|_{L^1} \leq \eta$, $|\int \nu_n^1 - \alpha| \leq \eta$, $|\int \nu_n^2 - (1 - \alpha)| \leq \eta$, and $\text{dist}(\text{supp } \nu_n^1, \text{supp } \nu_n^2) \rightarrow \infty$.

Each has a reading. In the first, all but a fraction η of the dissipation occurs within a time $R h_n$ of $t_n = h_n \tau_n$: a single pass, and π_0 is one atom. In the second no window of width $R h_n$ holds a fixed share. This is the case whenever the drag power is bounded uniformly in n , since then a window holds at most $2RC h_n/m_0$; the limit π_0 then has a bounded density and no atom. In the third the dissipation splits into pieces that separate on the fast scale: two passes a positive time apart, or a pass followed by a glide. Applying the lemma again to each piece, as in the construction of a profile decomposition, extracts the concentrated pieces one at a time; for a threshold δ the process stops after at most $\Delta\mathcal{E}/\delta$ steps, by Proposition B.4(ii), and leaves a remainder in which no window holds more than about δ . The remainder is small in the sense of the second alternative, in the norm $\sup_y \int_{y-R}^{y+R} |\cdot| d\tau$, and not in L^1 : on a glide it is most of the dissipation.

Remark B.5 (Which alternative occurs is decided by the corridor). Whether a family concentrates at all is a statement about how its corridor scales with ε , and belongs among the hypotheses. If the dynamic pressure and load limits are held fixed as $\varepsilon \rightarrow 0$, the drag power is bounded, every sequence vanishes, and the limit has no atoms: the trajectories are glide and phugoid, and the fast-scale structure is *oscillation*, for which the natural tool is averaging over the invariant measures of the fast flow [8]. Concentration requires loads that grow as $\varepsilon \rightarrow 0$, as in the classical matched-asymptotic analysis of skip entry at a fixed entry angle [23], where the layer has thickness $O(\varepsilon)$ in altitude. The physical value $\varepsilon \approx 1.1 \times 10^{-3}$, with entry angles of a few degrees, lies between the two: by Eq. (C.6) a pass that turns the flight path through 8° at a vertical lift-to-drag ratio of two dissipates about an eighth of the kinetic energy, and a computed pass lasts about a fifth of the slow time. The threshold in Eq. (2.2) is what makes the decomposition well posed at fixed ε without deciding the regime. Passes that dissipate more than δ are bubbles, and there are at most $\Delta\mathcal{E}/\delta$ of them whatever ε is; everything else is residual, and has to be bounded as oscillation.

Remark B.6 (The scale has to be chosen for each piece). Compactness at the scale h_n forces a single atom of π_0 , but vanishing at that scale does not exclude one: dissipation that concentrates at a coarser scale $h'_n \gg h_n$, $h'_n \rightarrow 0$, vanishes at the scale h_n and still produces an atom. Conversely a pass shorter than h_n is compact at that scale and its profile is a Dirac mass. The lemma classifies what happens at one scale; the width of each piece has to be found from the piece. That is the content of the limit-case form of the principle [22], in which compactness is recovered up to a translation and a dilation.

B.4 Profiles and their orthogonality

For an atom (t_j, a_j) of π_0 choose centers $t_n^j \rightarrow t_j$ and widths $h_n^j \rightarrow 0$ such that the windows $[t_n^j - R h_n^j, t_n^j + R h_n^j]$ capture a_j as $R \rightarrow \infty$, uniformly in n , with h_n^j the smallest scale at which this holds. The rescaled densities $s \mapsto h_n^j P_n(t_n^j + h_n^j s)$ are then tight on the line and converge weakly-* along a subsequence to a measure p^j of mass a_j , the *profile* of the pass. Two profiles $j \neq k$ are *orthogonal* in the sense of [26, (1.18)] when

$$\left| \log \frac{h_n^j}{h_n^k} \right| \rightarrow \infty \quad \text{or} \quad h_n^j = h_n^k \quad \text{and} \quad \frac{|t_n^j - t_n^k|}{h_n^j} \rightarrow \infty.$$

Atoms at distinct times are orthogonal by the second alternative, with nothing to prove. For orthogonal profiles the dissipation is additive,

$$\pi_n([0, T]) = \sum_{j=1}^J a_j + r_n^J([0, T]) + o(1) \quad (n \rightarrow \infty), \quad (\text{B.3})$$

with r_n^J the dissipation outside the J windows. This is the analogue of the energy identity [26, (1.25)], and it is immediate, because the dissipation is a measure and the windows are eventually disjoint. The analogue of the nonlinear statement [26, (1.26)]—that the solution is, up to a small error, the superposition of the profiles’ separate evolutions—is not immediate, and here it is not a superposition at all. An ordinary differential equation composes: the state leaving pass j is the datum of the arc that leads to pass $j + 1$. What replaces (1.26) is a factorization of the map from entry data to exit data through the individual pass maps and the outer flow, with a remainder that is explicit and uniform over the selections. That is objective 4.

B.5 Joint profiles where scales meet

Within a single pass the attitude and elastic motions are excited at their own scales, $h^{\text{att}} = \varepsilon_{\text{att}}$ and $h^{\text{ela}} = \varepsilon_{\text{ela}}$ in units of the slow time. They share the center of the pass, so the second alternative of (1.18) is not available and orthogonality has to come from the first: the ratio of the scales must diverge along the family. Where it does, the faster motion averages out over the slower. Where two periods remain comparable—where two bands cross inside the corridor—the corresponding profiles cannot be separated, and the decomposition carries one joint profile on the coupled subsystem. The check is pairwise and depends on the vehicle. For generic parametrizations the first structural period lies between 6 ms and 0.18 s and a pass lasts about 150 s; with the attitude period taken between 0.1 and 10 s, the atmospheric scale is separated from the other two by more than a decade, and the attitude and elastic scales meet only for a very slender, flexible body with a fast attitude loop. These are orders of magnitude, to be redone for any vehicle to which the result is applied.

B.6 What is not proved

1. *The singular continuous part.* Proposition B.4 allows $\pi_c \neq 0$: dissipation carried by a set of times of measure zero without atoms. Countably many passes that accumulate at a finite time, with summable dissipation—a sequence of decaying skips that ends in a glide, as the bounces of a ball end in rolling—are atoms and are already covered. A genuinely diffuse singular part would have to be excluded by a lower bound on the separation of passes in terms of the energy each dissipates. This step is open.

2. *Vanishing is glide.* That the part of π_0 with a density corresponds to motion near the glide balance, and how near, is the estimate of [Appendix C](#). It is an averaging estimate for a weakly damped oscillator, not a boundary-layer estimate, and it is not yet uniform over the selections.
3. *Compactness of the profiles over the family.* The profile p^j and the impulse set it generates depend on the selection. [Lemma B.3](#) bounds the diameter of the impulse set by $C_a a_j$; a description of the set itself is the inner problem of the classical theory, to be solved over the interval field.
4. *Composition.* The factorization that replaces (1.26), with its remainder, at each of the three scales and under interval disturbances.

C Skeleton residual and bubble budget

This appendix supports [Section 2.4](#). It gives the two constants of [Eq. \(2.3\)](#) in the form in which they presently exist, says what has been computed against them, and states the range in which each holds. The identities are checked by computer algebra, and the leading-order statements as limits of exact expressions, against the same definition of the field from which the numerical routines are generated; the comparisons with computation are verification tasks of the accompanying code, each with a failure criterion stated before it is run.

C.1 The planar glide

For spherical gravity and no rotation the altitude, speed and flight-path angle decouple from the rest of [Eqs. \(A.3\)](#) to [\(A.10\)](#):

$$\dot{r} = V \sin \gamma, \quad \dot{V} = -\frac{D}{m} - g \sin \gamma, \quad V\dot{\gamma} = \frac{L \cos \sigma}{m} - \left(g - \frac{V^2}{r}\right) \cos \gamma,$$

with $D/m = \rho V^2/(2\beta)$ for a ballistic coefficient β , an exponential density of scale height H_s , and $g = \mu/r^2$. Two abbreviations recur: the vertical lift-to-drag ratio and the net gravitational acceleration,

$$E_v = \frac{L}{D} \cos \sigma, \quad G = g - \frac{V^2}{r}.$$

On the equilibrium glide $\gamma = 0$ and $L \cos \sigma/m = G$, so $D/m = G/E_v$. As the speed bleeds off the glide altitude descends, and to leading order in $\varepsilon = H_s/R_e$ its rate is

$$\dot{h}_g = -\frac{2H_s g}{VE_v}. \quad (\text{C.1})$$

C.2 The phugoid

Linearized about a point of the glide with the coefficients frozen, the three equations have the characteristic polynomial $\lambda^3 + a_2\lambda^2 + a_1\lambda + a_0$, in which $a_2 = 2G/(E_v V)$ exactly, $H_s a_1 \rightarrow G$ and $a_0/a_1 \rightarrow -2V/(rE_v)$ as $H_s \rightarrow 0$. Its roots are one real root λ_s and a complex pair $-\varsigma \pm i\omega$, the *phugoid*: the oscillation of altitude against flight-path angle about the glide. The sum of the roots is $-a_2$, so $\varsigma = (a_2 + \lambda_s)/2$ exactly, and to leading order $\lambda_s = -a_0/a_1 = 2V/(rE_v)$. Hence

$$\omega^2 = \frac{G}{H_s}, \quad \varsigma = \frac{g}{VE_v}, \quad (\text{C.2})$$

each up to a relative correction of order ε . The real root is positive and is not an instability. It is the rate at which a frozen linearization registers that the glide itself is descending.

The two quantities in Eq. (C.2) scale differently, and everything in this appendix follows from that. The frequency grows like $\varepsilon^{-1/2}$. The damping rate does not contain H_s at all; by Eq. (C.1) it is $|\dot{h}_g|/(2H_s)$, half the rate at which the glide descends through scale heights. The damping ratio is therefore

$$\zeta = \frac{\varsigma}{\omega} = \frac{\sqrt{gR_e}}{VE_v\sqrt{1-V^2/(gr)}}\sqrt{\varepsilon}, \quad (\text{C.3})$$

of order $\sqrt{\varepsilon}$. For $L/D = 2$ at 6 km/s and the terrestrial scale height the period is about 265 s, the decay time $1/\zeta \approx 1250$ s and $\zeta \approx 0.034$: the oscillation lasts as long as the entry.

C.3 The residual constant

A trajectory started level on the glide altitude is not on the glide: the glide descends at the rate Eq. (C.1) and the trajectory trails it by one phugoid response time, ringing with an altitude amplitude $|\dot{h}_g|/\omega$. In scale heights that amplitude is $C_R\sqrt{\varepsilon}$ with

$$C_R = \frac{2\sqrt{gR_e}}{VE_v\sqrt{1-V^2/(gr)}}, \quad \text{so that} \quad C_R\sqrt{\varepsilon} = 2\zeta. \quad (\text{C.4})$$

The residual constant and the damping ratio are one number, read two ways.

The form has been measured rather than assumed. With the scale height varied from 2 to 30 km at fixed anchor density, the largest altitude deviation in scale heights scales as $\varepsilon^{0.49}$ and the largest flight-path angle as $\varepsilon^{0.98}$, so the altitude channel sets the bound. Against direct integration Eq. (C.4) is an upper bound in every case tried, tight to about four percent. The cases vary one quantity at a time about a generic lifting body at an angle of attack $\alpha = 16^\circ$, $V = 6$ km/s and zero bank, the angle of attack setting L/D :

case	L/D	computed	Eq. (C.4)	ratio
$\alpha = 10^\circ$	4.89	0.813	0.822	0.988
$\alpha = 16^\circ$	3.30	1.193	1.215	0.982
$\alpha = 25^\circ$	2.09	1.858	1.912	0.972
$\alpha = 35^\circ$	1.41	2.714	2.834	0.958
$V = 5.0$ km/s	3.30	1.207	1.236	0.977
$V = 6.8$ km/s	3.30	1.342	1.362	0.985
$\sigma = 30^\circ$	3.30	1.373	1.402	0.979
$\sigma = 60^\circ$	3.30	2.335	2.422	0.964

The formula bounds rather than equals because the trajectory rings: it tracks the envelope of the oscillation, and the computed supremum touches it from below.

C.4 Why this is not a settling rate

Equation (C.4) bounds the distance to the glide for a trajectory that starts on it. It does not say that other trajectories approach it, and by Eq. (C.3) they do so only slowly. The number of oscillations per e -folding of the amplitude is $1/(2\pi\zeta)$, which grows like $\varepsilon^{-1/2}$. In the limit the motion normal to the glide is an undamped oscillation on its own timescale. The glide balance is then not a normally hyperbolic slow manifold, the hypothesis under which a slow manifold persists and attracts [30, 43], and the regular-perturbation analysis of the glide notes the same thing from the other side: the exact solution does not approach the zeroth-order one [24].

The frozen rate is, moreover, not what either amplitude does. The coefficients drift as the vehicle slows, and with that drift retained the two amplitudes of the oscillation decay at different rates,

$$\frac{\dot{A}_h}{A_h} = -\frac{3}{2}\varsigma, \quad \frac{\dot{A}_\gamma}{A_\gamma} = -\frac{1}{2}\varsigma, \quad (\text{C.5})$$

for the altitude deviation and the flight-path angle respectively: the frequency rises as the speed falls, so the altitude swing closes faster and the angle swing slower. Their product, the action of the oscillator, decays at 2ς . Integration of the full equations confirms both rates to about one percent at the terrestrial scale height.

What excites the oscillation is at least as important as what damps it. A change of bank from σ_1 to σ_2 moves the glide altitude by $H_s \ln(\cos \sigma_2 / \cos \sigma_1)$, a fixed fraction of a scale height however thin the layer. Under bank modulation the altitude therefore departs from the instantaneous glide by an amount of order one in scale heights, not of order $\sqrt{\varepsilon}$, and so does a trajectory leaving a pass.

Two ways of closing the residual bound are consistent with this, and choosing between them is part of objective 5.

1. *Carry the amplitude.* The radius of the residual ball is the lag $C_{\mathcal{R}}\sqrt{\varepsilon}$ plus the ringing amplitude, bounded at its excitation and decaying by Eq. (C.5). The bound stays explicit and the skeleton stays the glide, at the cost of a residual that is not small immediately after a pass or a bank change.
2. *Average.* The phugoid amplitude is promoted to a slow coordinate of the skeleton and the phase is averaged out. The slow variables then follow the averaged inclusion and the fast ones converge in the statistical sense of [8]. The residual is the averaging error, which for a smooth field with a single fast phase is of order $\sqrt{\varepsilon}$ on the slow timescale; a rate that is uniform over the selections is what would have to be shown. The skeleton gains a dimension.

In the unstretched state variables the two descriptions agree on the order: an oscillation of fixed amplitude in scale heights has an altitude excursion of order εR_e and a flight-path-angle excursion of order $\sqrt{\varepsilon}$, so the occupation measures do concentrate on the glide as $\varepsilon \rightarrow 0$, at the rate $\sqrt{\varepsilon}$ set by the angle. What they do not do is concentrate at that rate in the metric of Eq. (2.3), in which altitude is measured in scale heights.

C.5 Range of validity

$C_{\mathcal{R}}$ diverges as $V^2 \rightarrow gr$. Near circular speed the lift required vanishes, the glide altitude climbs, and by Eq. (C.2) the phugoid slows until it is no longer fast: there is no separation of scales to exploit. The constant is therefore stated on a corridor bounded away from circular speed, and what lies above that bound is the regime of arcs and passes, to which the glide description does not apply. The same boundary appears independently in the attitude dynamics, where the dynamic pressure on the glide falls with the lift required and the trim loses stiffness, so that a floor on dynamic pressure belongs among the hypotheses of the corridor rather than among its consequences.

C.6 The bubble budget

By Lemma B.2 the specific energy is non-increasing along every trajectory of every selection, so a pass that dissipates at least δ can occur at most $K_\delta \leq \Delta\mathcal{E}/\delta$ times: this is Eq. (2.2). The dissipation of a pass is tied to how far it turns the flight path. Where the aerodynamic force dominates gravity,

$dV/V = -d\gamma/E_v$, and a pass that turns the flight path through $\Delta\gamma$ at constant E_v leaves with

$$\frac{V_{\text{out}}}{V_{\text{in}}} = e^{-\Delta\gamma/E_v}, \quad (\text{C.6})$$

the classical relation for a skip, in which $\Delta\gamma = 2\gamma_{\text{entry}}$ [28]. A pass entered at 4° with $E_v = 2$ therefore dissipates about thirteen percent of the kinetic energy, and an entry that sheds its energy in such passes makes about fourteen of them before the speed has fallen by a factor e . The count needs no heating correlation; the integrated heat load gives the same budget its physical reading, and the corridor constraint on it is carried separately (Remark A.11).

The width of a pass has been measured as well. A computed pass entered at 80 km, 6.5 km/s and -4° returns to its entry altitude after 145 s, which is 0.18 of the slow time $\sqrt{R_e/g}$; the phugoid period is $2\pi\sqrt{\varepsilon} = 0.21$ of it, and $\sqrt{\varepsilon} = 0.034$. The layer scales as $\sqrt{\varepsilon}$ and its width carries the factor 2π : an estimate that substitutes $\sqrt{\varepsilon}$ for the width understates it sixfold.

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